

At this moment, the group consists of four RTTs working part time for the study support group and the rest of their time on the Linacs. A trial coordinator keeps oversight of all the clinical trial related work, which includes Local and Central Datamanagement, Clinical Project Management, Statistics and the activities of the study support group.

The activities of the study support group vary per clinical trial. In general, the RTTs handle the Informed Consent Forms, send out, receive and process the study questionnaires and keep oversight of all study assessments per protocol. The study support group is the central contact point for all questions regarding the execution of a radiotherapy clinical trial for patients as well as radiation oncologists and other colleagues.

In the presentation, an overview will be given of the support provided by the RTTs for various clinical trials.

SP-0548 Real" Model in Radiation Therapist led research

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Abstract Text

Therapeutic radiographers (RT's) have a unique role in contributing and leading research. The role of the RT, along the patient pathway, gives rise to many opportunities to answer questions related to patient care and technology. It is the responsibility of the RT to investigate and answer these questions to improve treatment and patient experience. This talk will suggest and illustrate how to firstly, engage with research and become research aware, before moving onto becoming research active and eventually leading research. How to define a research question with suggestions of methods and tools which could be used to answer the question, will be presented. The issues of seeking and gaining funding will be discussed, in combination with defining and creating a research position and career.

Mini-Oral: CNS

MO-0549 Stereotactic radiosurgery in the management of uveal melanoma - treatment results of 594 patients

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Purpose or Objective

Various treatment options for uveal melanomas are available, including surgical resection, brachytherapy, proton therapy, fractionated radiotherapy, and stereotactic radiosurgery (SRS). However, extensive analyses on SRS are lacking. This study aims to assess the results of SRS for the treatment of uveal melanomas in a large patient cohort.

Materials and Methods

This single-institutional, retrospective study included patients with uveal melanomas treated with single-fraction robotic radiosurgery between 2005 and 2019. Eye retention, local control, and disease-specific survival rates were assessed using the Kaplan-Meier estimator. The impact of prescription dose, tumor size, and ciliary body involvement was evaluated with Cox proportional hazards models.

Results

A total of 594 patients met the inclusion criteria. In 22.7%, patients were stage I, 57.9% were stage II, 18.9% stage III, and 0.5% of patients were classified as stage IV. Median apical tumor height and base diameter were 5.8 and 11.4 mm, respectively. The mean follow-up was 41.7 months, and the median prescription dose was 21 Gy, with a median prescription isodose line of 70%. Local control rates after three and five years were 92.0% and 84.3%, respectively, for prescription doses of 21 and 22 Gy (448 patients) and 86.9% and 77.7%, respectively, when treated with a prescribed dose of 20 Gy or less (146 patients). The multivariable Cox regression model confirmed that a higher prescription dose had a significant positive impact on the local control (hazard ratio 0.73, $p=0.018$), whereas tumor size ($p=0.31$) and involvement of the ciliary body ($p=0.36$) had no effect. Eye retention was achieved in 89.9% and 81.0% after three and five years with 21 and 22 Gy, and 85.9% and 80.0% for 20 Gy or less. The multivariable Cox regression model demonstrated that only tumor size had a significant negative impact on eye retention (hazard ratio 1.52, $p=0.003$). In contrast, an effect of prescription dose ($p=0.28$) and involvement of the ciliary body ($p=0.89$) could not be confirmed. Disease-specific survival rates were 93.1% after three years, 89.8% after five years, and 87.8% after seven years. Tumor stage was associated with the disease-specific survival ($p=0.0001$). Serous retinal detachment was observed in 216 patients (36.4%). Secondary glaucoma occurred in 104 patients (17.5%), of whom 40 (38.5%) had to be enucleated due to intractable secondary glaucoma. Vitreous hemorrhage occurred in 74 patients (12.5%), and radiation retinopathy was detected in 72 patients (12.1%).

Conclusion

To the best of our knowledge, this analysis represents the most extensive patient cohort with uveal melanomas treated with SRS. Eye retention and local tumor control are achieved in the majority of cases, the latter mostly with prescription doses of at least 21 Gy. Retinal detachment is seen in about one-third of patients and should be anticipated during follow-up. These results suggest that SRS is not only a safe but also an effective treatment for uveal melanomas.

MO-0550 Correlations between cerebral cortical thickness, radiation dose and verbal memory performance

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